

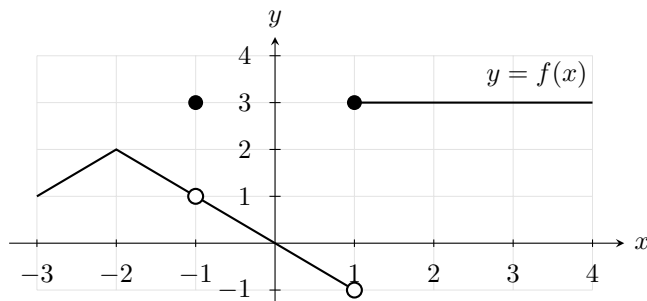
## Limits and Continuity: In-Class Practice

MATH 1334

Monday, October 5, 2026

Show your work and explain your reasoning. Give exact answers; no calculator is needed. Use radians. For a limit, give a real number,  $+\infty$ ,  $-\infty$ , or DNE, as appropriate. Use the review notes to decide what to revisit after your attempt.

1. Use the graph below. Open circles are excluded points; filled circles give the function values.



- (a) Find  $f(-1)$ ,  $\lim_{x \rightarrow -1^-} f(x)$ , and  $\lim_{x \rightarrow -1^+} f(x)$ . Is  $f$  continuous at  $-1$ ?
- (b) Find  $\lim_{x \rightarrow 1^-} f(x)$  and  $\lim_{x \rightarrow 1^+} f(x)$ . Can changing only  $f(1)$  make  $f$  continuous at  $1$ ? Explain.
- (c) Is  $f$  continuous at  $-2$ ? Explain why the corner does or does not matter.

*Review:* §§2.2–2.4: function values, one-sided limits, continuity, and discontinuities.

2. Let  $H(x) = \frac{-2}{(x+2)(x-1)^2}$ .

- (a) Find the left-hand, right-hand, and two-sided limits at  $x = -2$ .
- (b) Find the left-hand, right-hand, and two-sided limits at  $x = 1$ .
- (c) Give all vertical asymptotes. Explain how the powers of the denominator's factors affect the signs in (a) and (b).

*Review:* §2.2: sign analysis, infinite limits, and vertical asymptotes.

**3.** Evaluate each limit. Choose a method and show the steps that justify it.

(a)  $\lim_{x \rightarrow 2} \frac{x^2 + 3}{x + 1}$

(b)  $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x^2 - 9}$

(c)  $\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - \sqrt{4-x}}{x}$

(d)  $\lim_{x \rightarrow 0} \frac{\frac{1}{1+x} - \frac{1}{1-x}}{x}$

*Review:* §2.3: substitution, factoring, conjugates, and common denominators.

**4.** Evaluate each limit. Show the identity or fundamental limit you use.

(a)  $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(2x)}$

(b)  $\lim_{x \rightarrow 0} \frac{\tan(2x)}{3x}$

*Review:* §2.3: the fundamental sine limit, matching angles, and the identity  $\tan u = \sin u / \cos u$ .

5. Find all values of  $a$ ,  $b$ , and  $c$  that make  $G$  continuous on  $\mathbb{R}$ . Write the continuity conditions at both 1 and 3, including the function value at each point.

$$G(x) = \begin{cases} x^2 + 1, & x < 1, \\ ax + b, & 1 \leq x < 3, \\ c, & x = 3, \\ 2x, & x > 3. \end{cases}$$

*Review:* §2.4: the three continuity conditions and matching pieces at their endpoints.

6. Evaluate the following arguments. Explain what is correct or incorrect.

(a) A student says, “Substitution gives  $0/0$ , so the limit does not exist.” Give a specific example showing why this conclusion is not justified.

(b) Suppose  $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 3$ , but  $f(2) = -1$ . A student says, “The one-sided limits agree, so  $f$  is continuous at 2.” Identify the failed continuity condition and explain how to repair it.

*Review:* §§2.3–2.4: indeterminate forms and the difference between a limit and a function value.

7. Sketch two continuous functions on  $[-2, 2]$ , each with endpoint values  $f(-2) = f(2) = 3$ : one with no zeros, and one with at least two zeros. What do positive endpoint values alone tell you about whether a continuous function has a zero inside the interval?

*Review:* §2.4: what the Intermediate Value Theorem guarantees about zeros.

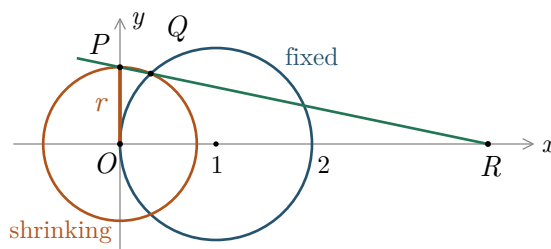
8. For  $x \neq 0$ , let  $q(x) = 2 + x \sin(1/x)$ . Find  $\lim_{x \rightarrow 0} q(x)$  using the squeeze theorem. Give lower and upper bounds valid for both positive and negative  $x$ , and show their limits.

*Review:* §2.3: the squeeze theorem and bounds that hold on both sides of the limiting input.

9. Show that  $\cos x = x$  has at least one solution in  $(0, \pi/2)$ . Name the function to which you apply the Intermediate Value Theorem, verify its hypotheses, and state its conclusion. Does the IVT alone prove uniqueness?

*Review:* §2.4: the Intermediate Value Theorem, its hypotheses, and existence versus uniqueness.

10. **Where does the intercept go?** Keep the circle  $(x-1)^2 + y^2 = 1$  fixed while the circle  $x^2 + y^2 = r^2$  shrinks toward  $O = (0, 0)$ , with  $0 < r < 1$ . Let  $P = (0, r)$ , let  $Q$  be the upper intersection of the circles, and let  $R = (a(r), 0)$  be the  $x$ -intercept of the line through  $P$  and  $Q$ .



(a) Sketch a smaller circle and its corresponding line  $PQ$ . Predict where  $R$  goes as  $r \rightarrow 0^+$ . Does it have to follow  $P$  and  $Q$  toward the origin?

(b) Find the coordinates of  $Q$  in terms of  $r$ , then find  $a(r)$ . *Hint: subtract the two circle equations before solving for the positive  $y$ -coordinate.*

(c) Find  $\lim_{r \rightarrow 0^+} a(r)$  and justify your answer algebraically. Compare with your prediction. Why can you not determine the intercept by setting  $r = 0$  in the geometric construction at the outset?

*Review:* Circle equations, slopes, and intercepts; §2.3: conjugates and indeterminate forms.